

An Optimal Guidance Strategy for the Defense of a Non-Maneuverable Target in 3-Dimensions

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Abstract: An engagement scenario involving the defense of a non-maneuverable agent, called the Target, is considered. A pursuing agent, engages the non-maneuverable Target while a defending agent, which aims at intercepting the Pursuer in order to protect the Target, is considered in 3-Dimensions. A zero-sum differential game is formulated where the objective of the Defender is to capture the Pursuer and maximize the terminal Pursuer-Target range at final time; while the Pursuer aims at minimizing the terminal Pursuer-Target range at final time. The terminal time is not fixed and is determined by the interception of the Pursuer by the Defender. The saddle point state-feedback strategies for the Pursuer and the Defender are obtained using a gradient-based minimization routine. A comparison of the saddle point strategy against a heuristic approach, Pure Pursuit, is conducted. The comparison demonstrates that poorer performance is achieved by players who deviate from the optimal saddle point strategy; and, as a result, the saddle point strategies outperform the heuristic method.

1 Introduction

The modeling and analysis of pursuit-evasion scenarios involving multiple agents represent an important area of interest for the military, aerospace, and controls community. The utilization of differential games to accomplish the analysis of pursuit-evasion was originally outlined by Isaacs [1]. While Isaacs' notable work considered pursuit-evasion in the plane, others extended his work to include pursuit-evasion in 3-Dimensions. One example of a closed form solution to a three-dimensional missile-aircraft pursuit-evasion game was posed by Shinar and Gutman [2]. Shinar also investigated a realistic pursuit-evasion engagement involving missile-air and air-air scenarios using variational methods [3]. Hillberg and Chalmers investigated a pursuit-evasion game between two realistic aircraft in a dogfight using distance as a metric for optimization. They took into account physical aircraft limits such as structural and stall limits [4]. Greenwood designed a realistic differential game in 3-Dimensions by modeling fighter aircraft [5]. Greenwood used dynamics of two fighter aircraft in space and even considered firing envelopes as part of this barrier analysis. Imado and Kuroda proposed a differential game involving a missile-aircraft pursuit-evasion engagement where a nonlinear miss-distance was used as a payoff function [6]. Shinar, Glizer, and Turetsky, investigated a pursuit-evasion game where the dynamics of the pursuer could be changed during the pursuit a finite number of times similar to multi-stage missiles [7]. In their paper [8], the Target had the ability to change its dynamics during the engagement a finite number of times. Merz proposed a problem of pursuit or evasion selection if both agents were tactical and role assignment had not yet happened [9]. Related to dog-fights and aerial combat, Merz's concern was with the differential game assignment and the outcome of the engagement.

The consideration of multiple agents engaged in pursuit and defense of a moving or stationary target has been considered in the past. Introduced by Boyell [10, 11], the defense of a ship from an incoming torpedo using a counter-weapon was described. The two-players one evader game of Cardaliaguet [12] is an early version of target defense. In the paper, a two-player differential game is played wherein one of the players wants the state of the system to reach an evader, while the other player wants the state of the system to

avoid this evader. Yamasaki and Balakrishnan proposed the defense of active Target by the launching of a defensive missile [13, 14]. In their work, they proposed a closed-loop feedback control law to defend a target aircraft successfully. Using simple motion, Pachter, Garcia, and Casbeer [15] consider a zero-sum three-agent pursuit-evasion differential game. Additionally they considered a case in 2-Dimensions when the Defender was faster than both the Pursuer and the Target [16].

Previous works have considered target defense but have restricted the game to the 2-Dimensional plane [16–20]. Rather than consider the defense of a non-maneuvering target in 2-Dimensions, we consider the problem in 3-Dimensions. By considering the engagement in 3-Dimensions, we are able to introduce the concept of altitude to the differential game described in [18]. The inclusion of altitude and dynamics in 3-Dimensions, are in-general, more realistic for the modeling of aerial engagements. Utilizing geometry of Apollonius, we are able to not only set-up the differential game, but solve its numerically.

One common issue with posing differential games in 3-Dimensions when compared to those in 2-Dimensions is the tractability of analytic solutions. The added spatial dimension begs the use of spherical coordinates as well as additional controls. In the 2-Dimensional problem one control for each agent is considered (agent headings). However, in 3-Dimensions, each agent has two controls, one for azimuth and one for elevation. In total, the dynamics become more nonlinear, more controls are required to pose and solve 3-Dimensional differential games, and as a result, tractable solutions are challenging to find.

In this work we consider an engagement scenario involving the defense of a non-maneuverable agent, called the Target. A Pursuer, engages the non-maneuverable Target while the Defender, which aims at intercepting the Pursuer in order to protect the Target, is considered in 3-Dimensions. By posing a scenario of a non-maneuvering Target, we consider the differential game which develops between two players: Pursuer and Defender. Preventing the Target from maneuvering, a differential game is posed where it is the goal of the Defender to capture the Pursuer as far from the Target as possible. A non-maneuvering Target would be representative of aircraft which

are very slow to maneuver or are set on a fixed course and heading such as a landing approach or take-off.

We consider a perfect information game where each agent has access to the state of all players. We do not model sensor measurements, state uncertainty, or stochastics in this brief. Furthermore, we assume a purely kinematic model for each agent, neglecting gravity and mass. The control for the agents heading and climb angles are unbounded. Similar to other work [21–24], the headings of the agents are the controlled parameter. Utilizing this simple kinematic model, the motion of the agents are represented by a set of non-linear ordinary differential equations, representative of aircraft in 3-Dimensions.

Assuming a game of kind (successful defense of the Target by the Defender), a zero-sum differential game is formulated where the objective of the Defender is to intercept the Pursuer and to maximize the terminal Pursuer-Target range at final time; while the Pursuer aims at minimizing the terminal Pursuer-Target range at final time. The terminal time is not fixed and is determined by the interception of the Pursuer by the Defender. The saddle point state-feedback strategies for the Pursuer and the Defender are obtained using simple geometry; these strategies are compared to the heuristic approach of Pure Pursuit.

In this work we determine the saddle point strategies of the Pursuer and Defender. We also show that these strategies are very different from the heuristic strategy of Pure Pursuit. Specifically, we show that if one of the players implements the optimal strategy derived in the paper while the opponent utilizes Pure Pursuit, then the first player will see the value of the game shift in its favor: i.e. agents benefit from the opponent playing a non-optimal heuristic strategy.

This paper describes the setup and solution to the differential game as well as an analysis and description of a numerical simulation in MATLAB. In Section 2 we begin by setting the equations-of-motion and defining the Differential Game Hamiltonian. Next, in Section 3, the geometry which describes the optimal policies of the Defender and Pursuer are presented. Using the derived optimal policies, we conduct three different simulations in Section 4. In one simulation, both Pursuer and Defender implement the optimal policies described in Section 3. Next, we simulate a suboptimal policy for the Pursuer and optimal policy for the Defender. Finally we conduct the simulation where the Pursuer chooses an optimal policy and the Defender selects a suboptimal one. Following the mathematical contributions, numerical simulations of the optimal saddle point strategies are presented with discussion. Finally, in Section 5 we draw conclusions about the analysis and summarize the work described in this paper.

2 Differential Game

Considering the defense of a non-maneuverable agent, we consider three agents with constant velocity in 3-Dimensions. We define the constant velocities of the Pursuer, Target, and Defender as velocities v_P , v_T , and v_D respectively. The complete state of the game is defined by $\mathbf{x} = [x_P, y_P, z_P, x_D, y_D, z_D, x_T, y_T, z_T] \in \mathbb{R}^9$. The Pursuer's control, $\mathbf{u}_P$, is defined by instantaneous heading angle, ψ_P , and instantaneous elevation angle, θ_P : $\mathbf{u}_P = \{\psi_P, \theta_P\}$. Similar to the Pursuer, the Defender's control, $\mathbf{u}_D$, is composed of his instantaneous heading ψ_D and elevation angle, θ_D : $\mathbf{u}_D = \{\psi_D, \theta_D\}$. The dynamics, $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mathbf{u}_P, \mathbf{u}_D)$, of the control system are defined by the ordinary differential equation in (1). In this model, we assume that agents move with simple motion are able to turn instantaneously.

$$\begin{bmatrix} \dot{x}_P \\ \dot{y}_P \\ \dot{z}_P \\ \dot{x}_T \\ \dot{y}_T \\ \dot{z}_T \\ \dot{x}_D \\ \dot{y}_D \\ \dot{z}_D \end{bmatrix} = \begin{bmatrix} v_P \cos \theta_P \cos \psi_P \\ v_P \cos \theta_P \sin \psi_P \\ v_P \sin \theta_P \\ v_T \cos \theta_T \cos \psi_T \\ v_T \cos \theta_T \sin \psi_T \\ v_T \sin \theta_T \\ v_D \cos \theta_D \cos \psi_D \\ v_D \cos \theta_D \sin \psi_D \\ v_D \sin \theta_D \end{bmatrix} \quad (1)$$

The domain of the control variables for each of the agents are: $\psi_P, \psi_D \in [0, \pi]$ and $\theta_P, \theta_D \in [-\pi, \pi]$. While the Target's heading and flight path angle is known and constant for the entire engagement, the dynamics of the Target are included because they effect the cost/payoff of the differential game. Including the Target dynamics is necessary due to the fact that this is a free-terminal time differential game and the terminal position of the Target is unknown in advance, even when its heading and flight path angle angles are known and constant.

Define the speed ratio parameter

$$\alpha = v_T/v_P \quad (2)$$

In general, the Pursuer is faster than the Target, so $\alpha < 1$. Also define the speed ratio

$$\gamma = v_P/v_D \quad (3)$$

For this paper we consider the case where the Pursuer is slower than the Defender, so $\gamma < 1$. Thus, we consider the engagement where

$$v_D > v_P > v_T \quad (4)$$

In this paper we consider the provided a-priori initial state of the three agents (Pursuer, Defender, and Target) to be, $\mathbf{x}_0$

$$\mathbf{x}_0 = [x_{P_0}, y_{P_0}, z_{P_0}, x_{D_0}, y_{D_0}, z_{D_0}, x_{T_0}, y_{T_0}, z_{T_0}] \quad (5)$$

Further, we consider point-capture. Where the separation distance (P, D) has to become zero in order for the fast Defender to intercept the Pursuer. Also, we assume that given the initial states belong to the escape set, denoted by $\mathcal{R}_e \subset \mathbb{R}^9$. Having the solution to a Game of Kind, where the successful defense of the Target by the Defender is possible, we are concerned with the Game of Degree, how well defense is conducted in the event that the Defender is capable of capturing the Pursuer before the Pursuer reaches the Target. The Defender wants to maximize the terminal P - E distance while the Pursuer works to minimize the same. The game of degree is played where the states of the system does not belong to $\mathcal{R}_e$ and capture of the Target by the Pursuer is possible. This is analogous to the active defense differential game of maneuverable aircraft in [25]. There exists a strategy for the Defender, $\mathbf{u}_D$, such that the Defender captures the Pursuer before the Pursuer is capable of capturing the Target, T . The game of capture, one wherein the Pursuer captures the Target before the Defender can capture the Pursuer, is not considered in this paper and is left for future investigation. The terminal manifold which describes the point capture of the Pursuer by the Defender is

$$\mathcal{C} = \left\{ \mathbf{x} \mid (x_P - x_D)^2 + (y_P - y_D)^2 + (z_P - z_D)^2 = 0 \right\} \quad (6)$$

The terminal time, t_f is the instant where the state reaches the terminal, $\mathcal{C}$, specified in (6). The terminal state is

$$\mathbf{x}_f = [x_{P_f}, y_{P_f}, z_{P_f}, x_{D_f}, y_{D_f}, z_{D_f}, x_{T_f}, y_{T_f}, z_{T_f}] \quad (7)$$

The terminal cost/payoff functional is

$$J(\mathbf{u}_P, \mathbf{u}_D, \mathbf{x}_0) = \Phi(\mathbf{x}_f) \quad (8)$$

where

$$\Phi(\mathbf{x}_f) = (x_{P_f} - x_{E_f})^2 + (y_{P_f} - y_{E_f})^2 + (z_{P_f} - z_{E_f})^2 \quad (9)$$

The terminal cost/payoff functional depends only upon the terminal state of the active target defense differential game (ATDDG); its value is given by

$$V(\mathbf{x}_0) = \min_{\mathbf{u}_P(\cdot)} \max_{\mathbf{u}_D(\cdot)} J(\mathbf{u}_P, \mathbf{u}_D, \mathbf{x}_0) \quad (10)$$

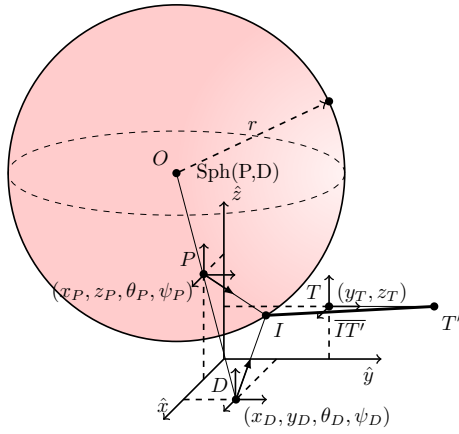


Fig. 1: Interception of Pursuer by Defender at Terminal Time

The optimization is conducted subject to the dynamics (1) and trajectories with terminal state (7), where $\mathbf{u}_P$ and $\mathbf{u}_D$ are the players' state feedback strategies.

The co-states are

$$\lambda^T = [\lambda_{x_P}, \lambda_{y_P}, \lambda_{z_P}, \lambda_{x_E}, \lambda_{y_E}, \lambda_{z_E}, \lambda_{x_D}, \lambda_{y_D}, \lambda_{z_D}] \in \mathbb{R}^9 \quad (11)$$

In this case, where the cost/payoff functional is terminal; the Hamiltonian of the differential game is simply the inner product of the dynamics with the co-states

$$\mathcal{H} = \langle \lambda^T, \mathbf{f}(\mathbf{x}, \mathbf{u}_P, \mathbf{u}_D) \rangle \quad (12)$$

Utilizing the speed ratios in (2) and (3), the velocities of the agents are normalized with respect to the Defender to obtain the Hamiltonian

$$\begin{aligned} \mathcal{H} = & \lambda_{x_P} \gamma \cos \theta_P \cos \psi_P + \lambda_{y_P} \gamma \cos \theta_P \sin \psi_P \\ & + \lambda_{z_P} \gamma \sin \theta_P + \lambda_{x_E} \alpha \gamma \cos \theta_E \cos \psi_E \\ & + \lambda_{y_E} \alpha \gamma \cos \theta_E \sin \psi_E + \lambda_{z_E} \alpha \gamma \sin \theta_E \\ & + \lambda_{x_D} \cos \theta_D \cos \psi_D + \lambda_{y_D} \cos \theta_D \sin \psi_D \\ & + \lambda_{z_D} \sin \theta_D \end{aligned} \quad (13)$$

Theorem 1. Consider the defense of a non-maneuverable agent differential game in 3-Dimensions (1)-(10). The chosen heading and flight path angles of the Pursuer and Defender under optimal play are constant and their optimal trajectories are straight lines.

Proof: See Appendix A. $\square$

3 Strategies

From Theorem 1, the optimal trajectories are straight lines; using the Cartesian coordinate frame to describe a 3-Dimensional pursuit-evasion differential game we can investigate optimal strategies for the three-agent scenario. We simplify the governing equations by setting the origin of the Cartesian space as shown in Fig. 1. By locating each agent on an orthogonal plane and orienting the y-axis parallel to the Target, we can eliminate some variables during our analysis.

Using this coordinate frame, we are able to reduce the state space by three variables as well as fully describe the intersection sphere of the Defender, D , and Pursuer, P .

The two-norm distance between the Pursuer and the Defender is

$$d = \sqrt{(x_P - x_D)^2 + y_D^2 + z_P^2} \quad (14)$$

By setting the origin of the space as described in Fig. 1, the variables $x_E = y_P = z_D = 0$. In order to fully define the intersection sphere, we need to locate the sphere's origin and specify its radius. We know the origin is located collinear with the points: P , and D , the position of the Pursuer and Defender. If we define the ray passing through P and D as $\hat{x}'$, then the origin's location can be easily obtained. Let us designate the unit vector for the ray $\overrightarrow{PD}$ as $\hat{x}'$

$$\hat{x}' = \frac{1}{d} ((x_D - x_P)\hat{i} + (y_D)\hat{j} + (-z_P)\hat{k}) \quad (15)$$

The origin, O is collinear with $\hat{x}'$ and it lies in the opposite direction at a distance

$$\overline{OP} = \frac{\gamma^2 d}{1 - \gamma^2} \quad (16)$$

Simply reversing the unit vector, $\hat{x}'$ and multiplying by the magnitude of $\overline{OP}$, we obtain the location of the origin to be

$$O = -\overline{OP}\hat{x}' + P \quad (17)$$

More explicitly, the point

$$O = \frac{1}{1 - \gamma^2} \begin{bmatrix} x_P - x_D \gamma^2 \\ -y_D \gamma^2 \\ z_P \end{bmatrix} \quad (18)$$

We specify the radius of the sphere to be that of the radius of the Apollonius circle whose P and D foci are at a distance, d , and the speed ratio is γ .

$$r = \frac{\gamma d}{1 - \gamma^2} \quad (19)$$

3.1 Agent Headings

By Theorem 1, the optimal headings of the players are constant. In order to obtain the optimal strategies of the Pursuer and the Defender we first select a point, I , on the surface of the Intersection Sphere to serve as an intersection point. This intersection point determines the location at which the Defender captures the Pursuer. Note that the optimal interception point is not guessed by the Defender, rather it is obtained by solving the differential game. Utilizing spherical coordinates, (r, θ_O, ψ_O) , we begin by locating the intersection point relative to the Intersection Sphere's origin

$$\begin{aligned} x_I &= r \cos \theta_O \cos \psi_O + x_O \\ y_I &= r \cos \theta_O \sin \psi_O + y_O \\ z_I &= r \sin \theta_O + z_O \end{aligned} \quad (20)$$

From (20) we obtain the corresponding Pursuer and Defender control in order to take a straight line path to the intersection point.

3.1.1 Pursuer Heading: The pursuer strategy is specified by the heading, ψ_P , and flight path angle, θ_P , which leads the Pursuer to reach a given intersection point on the Intersection Sphere

$$\mathbf{u}_P = \{(\theta_P, \psi_P) | \theta_P \in [-\pi, \pi], \psi_P \in [0, \pi]\} \quad (21)$$

The intersection point, I , using Cartesian coordinates, as a function of the Pursuer's heading and flight path angle is

$$\begin{aligned} x_I &= \overline{PI} \cos \theta_P \cos \psi_P + x_P \\ y_I &= \overline{PI} \cos \theta_P \sin \psi_P \\ z_I &= \overline{PI} \sin \theta_P + z_P \end{aligned} \quad (22)$$

The distance, $\overline{PI}$ is

$$\overline{PI} = \sqrt{(x_P - x_I)^2 + y_I^2 + (z_P - z_I)^2} \quad (23)$$

Therefore we obtain the Pursuer's strategy $\mathbf{u}_P$ as a function of pursuer coordinates and intersection point coordinates: $(x_P, z_P, x_I, y_I, z_I)$

$$\theta_P = \sin^{-1}((z_I - z_P) / \overline{PI}) \quad (24)$$

$$\psi_P = \sin^{-1}\left(y_I / \left(\overline{PI} \sqrt{1 - \frac{(z_I - z_P)^2}{|\overline{PI}|^2}}\right)\right) \quad (25)$$

3.1.2 Defender Heading: Knowing the intersection point, I , we also obtain the corresponding strategy $\mathbf{u}_D$ of the Defender. The Intersection point of the Pursuer by the Defender in spherical coordinates is

$$\mathbf{u}_D = \{(\theta_D, \psi_D) | \theta_D \in [-\pi, \pi], \psi_D \in [0, \pi]\} \quad (26)$$

The intersection point, I , can be described in Cartesian coordinate frame as

$$\begin{aligned} x_I &= \overline{DI} \cos \theta_P \cos \psi_P + x_D \\ y_I &= \overline{DI} \cos \theta_P \sin \psi_P + y_D \\ z_I &= \overline{DI} \sin \theta_P \end{aligned} \quad (27)$$

The distance, $\overline{DI}$, is

$$\overline{DI} = \sqrt{(x_D - x_I)^2 + (y_D - y_I)^2 + z_I^2} \quad (28)$$

Therefore we write the Defender's strategy, $\mathbf{u}_D$ as a function of the defender coordinates and intersection point coordinates $(x_D, y_D, x_I, y_I, z_I)$

$$\theta_D = \sin^{-1}(z_I / \overline{DI}) \quad (29)$$

$$\psi_D = \sin^{-1}\left((y_I - y_D) / \left(\overline{DI} \sqrt{1 - \frac{z_I^2}{|\overline{DI}|^2}}\right)\right) \quad (30)$$

3.2 Optimal Intersection Point

The agent headings (24), (25), (29), (30) are functions of a chosen intersection point, I , located on the Intersection Sphere. We perform the optimization to minimize the desired objective functional (8)

$$(\theta_O^*, \psi_O^*) = \arg \min_{\theta_O, \psi_O} \Phi(\mathbf{x}_f) \quad (31)$$

where the final time is computed as follows

$$t_f = \frac{\overline{PI}}{v_P} = \frac{|\overline{EE'}|}{v_E} = \frac{|\overline{DI}|}{v_D} \quad (32)$$

The location of the Target at the time of interception, t_f is found utilizing the speed ratio α as

$$|\overline{EE'}| = \frac{v_E}{v_P} |\overline{PI}| = \alpha |\overline{PI}| \quad (33)$$

We use a gradient based method to obtain the optimal solution are utilized to find the optimal strategies for the Pursuer and Defender.

3.3 Special Case: 2-Dimensional Engagement

In the event that all three agents have zero-vertical separation, e.g., $z_P = z_E = z_D = 0$. The engagement amounts to a 2-Dimensional problem described in [18, 19], and the analysis from [18] can be used to compute the optimal solutions to the differential game analytically.

4 Simulation

Consider an example engagement where the agent positions are: $P_0 = (5, 0, 10)$, $E_0 = (0, 10, 5)$, $D_0 = (10, 5, 0)$, outlined in Table 1: Simulation Initial Conditions. Furthermore, we consider the agent velocities: $v_P = 0.9$, $v_E = 0.8$, and $v_D = 1.0$. We set the Target fixed heading and climb angle to be: $\psi_E = \pi/2$ and $\theta_E = 0$.

4.1 Example 1: Optimal Pursuer and Optimal Defender

Fig. 2 shows the optimal trajectories for the scenario where each of the players, P and D , implement the saddle point strategies outline from the solution to the differential game. The value of the game is $V(\mathbf{u}_P^*, \mathbf{u}_D^*, \mathbf{x}_0) = 9.9764$; summarized in Table 2: Simulation Results.

Table 1 Simulation Initial Conditions

Agent	Coordinate	Location
Pursuer	x_P	5.0
Pursuer	y_P	0.0
Pursuer	z_P	10.0
Target	x_T	0.0
Target	y_T	10.0
Target	z_T	5.0
Defender	x_D	10.0
Defender	y_D	5.0
Defender	z_D	0.0

Note that the optimal saddle point strategies are computed at every time step and implemented for both the Defender and Pursuer. Utilizing the positions of all agents, the differential game is solved wherein the positions of all agents are used to obtain the appropriate instantaneous headings of the Pursuer and Defender. The solution of the defense of the non-maneuverable agent possess the invariance property, where the interception point and headings of all players are constant under the saddle point solution to the differential game. If, however, any of the agents maneuvers in a way not prescribed by the optimal strategies of the differential game, then the optimal headings computed will not be constant. The remaining examples in the paper implement heuristic Pure Pursuit strategies for the Defender and Pursuer showing that performance of the agents only degrades with variation from the saddle point strategies.

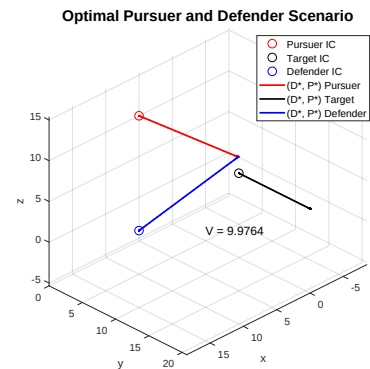


Fig. 2: Optimal Pursuer and Defender Engagement Scenario

4.2 Example 2: Optimal Defender and Heuristic Pursuer

Consider the case where the Pursuer does not follow the optimal saddle point strategy outlined in this paper; rather, it implements a Pure Pursuit strategy, where it's heading and climb angle are chosen to point towards the Target at each time step. The equations utilized

for computing the control for the suboptimal-pursuer are

$$\theta_P = \tan^{-1} \left(\frac{z_T - z_P}{\sqrt{(x_T - x_P)^2 + (y_T - y_P)^2}} \right) \quad (34)$$

$$\psi_P = \tan^{-1} \left(\frac{y_T - y_P}{x_E - x_P} \right) \quad (35)$$

While the Pursuer implements the Pure-Pursuit guidance strategy defined by (34) and (35), the Defender implements the optimal solution to the differential game: (29) and (30). The resulting simulation can be seen in Fig. 3. As a result of the sub-optimal strategy performed by the Pursuer, the Defender is able to capture the Pursuer sooner and as a result the terminal distance is larger: $V(\mathbf{u}_P, \mathbf{u}_D^*, \mathbf{x}_0) = 10.1936$. Table 2 compares the performance of the suboptimal Pursuer strategy against that of an optimal one.

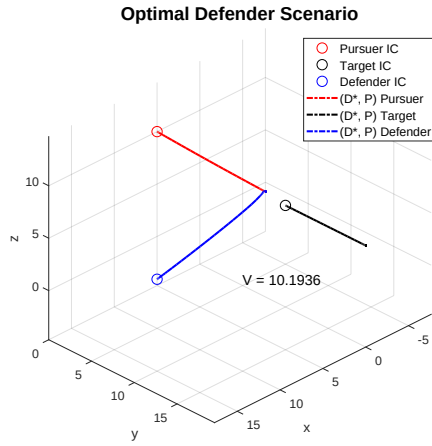


Fig. 3: Optimal Defender Engagement Scenario

4.3 Example 3: Heuristic Defender and Optimal Pursuer

Finally, consider the opposite case, wherein the Pursuer follows the optimal strategy derived in this paper while the Defender implements the sub-optimal Pure Pursuit strategy. The Pure Pursuit headings are obtained by pointing the Defender towards the Pursuer, as seen in the following equations

$$\theta_D = \tan^{-1} \left(\frac{z_P - z_D}{\sqrt{(x_P - x_D)^2 + (y_P - y_D)^2}} \right) \quad (36)$$

$$\psi_D = \tan^{-1} \left(\frac{y_P - y_D}{x_P - x_D} \right) \quad (37)$$

The Pursuer, implements the optimal strategy from (25) and (24); the resulting trajectories are shown in Fig. 4. The terminal separation at final time is, as a result of the suboptimal Defender strategy is less than when the saddle point strategy is implemented. The value function, describing this terminal separation is: $V(\mathbf{u}_P^*, \mathbf{u}_D, \mathbf{x}_0) = 9.2166$, showing that deviation from the solution to the differential game results in poorer performance for the Defender.

Highlighted in Table 2, we observe a comparison of the three simulations which demonstrates the characteristic of the solution to a differential game. First, both the Pursuer and the Defender implement the optimal saddle point strategies and observe the terminal Pursuer-Target range is 9.9764. When the Defender implements Pure Pursuit and the Pursuer implements the optimal saddle point strategy, the value function is lower and the Pursuer gets closer to the Target at final time. The resulting range is 9.2166. When the Pursuer implements Pure Pursuit and the Defender implements the

Table 2 Simulation Results

Defender Strategy	Pursuer Strategy	Value Function
Optimal	Optimal	9.9764
Pure Pursuit	Optimal	9.2166
Optimal	Pure Pursuit	10.1936

optimal saddle point strategy, the terminal range between the Pursuer and Target is 10.1936. We observe, upon implementing the heuristic sub-optimal policy of Pure Pursuit, performance is reduced for that agent. This demonstrates that deviating from the optimal saddle point strategy only reduces performance for that agent, a characteristic of differential games [1].

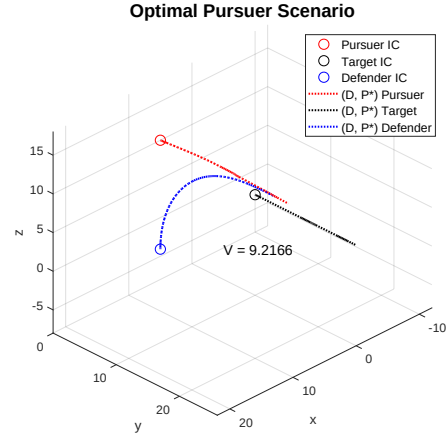


Fig. 4: Optimal Pursuer Engagement Scenario

4.4 Search Performance

Utilizing the MATLAB function `FMINUNC()`, we compute the optimal headings directly. A plot of the entire space is presented in Fig. 5.

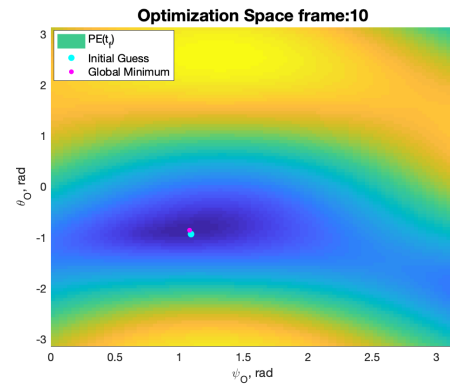


Fig. 5: Domain of Headings and Evaluation of Pursuer-Target Range at t_f

In Fig. 5 we observe the space in the example is locally convex. By seeding the optimization search with a guess for the optimal heading angles, we can converge fairly quickly with fewer functional evaluations than brute-force searching the entire domain for the minimum. Using the Matlab Function, `FMINUNC()`, we implement a gradient based search for the optimal headings described in (31). Since the gradient based search requires an initial guess for finding the local minimum, we provide the headings ψ_O and θ_O described

by the vector from the origin of the interception sphere to the initial location of the Target as an initial seed. Using the Quasi-Newton gradient based search algorithm, we conduct a search for the optimal interception point described by ψ_O^* and θ_O^* . Using an initial guess of the headings from the Origin to the Target, the guess is relatively close to the final converged global minimum, and optimal solutions are computed generally within 20-30 functional evaluations. If the initial guess were to start very far away from the optimal solution, then a gradient based solver would not converge on the global solution. As illustrated in Fig. 5, initial guesses are not far from the global minimum, and in-general, convergence to the global minimum is achieved.

Utilizing MATLAB, the gradient based search takes an initial guess for the location of the interception point located on the interception sphere. An initial guess of ψ_O and θ_O which points from the origin, O , to the initial location of the Evader, E , is used. This point on the interception sphere is shown in light-blue in Fig. 5. Upon finding the optimal interception point, the headings of the Pursuer and Defender are then implemented using (24-25, 29-30). Conducting the gradient based search outperforms that of a brute force search. Rather than computing the terminal cost for every point on the sphere, described by the color gradient in Fig. 5, the use of gradient based techniques require fewer functional evaluations and is more efficient than Brute Force search.

5 Conclusions

We have posed a 3-Dimensional pursuit-evasion differential game with a 9-dimensional state space. The differential game of a non-maneuvering Target is engaged by an attacking agent, called the Pursuer. In order to defend the Target, a Defender aims at intercepting the Pursuer before it can reach the Target. Utilizing an intersection sphere we were able to formulate the Game of Degree assuming the Game of Kind: the successful defense of a non-maneuvering Target has been solved. This interception sphere described the locus of points of possible interceptions of the Pursuer by the Defender. Leveraging the geometric concept of an Apollonius circle, we were able to transform the search of saddle point strategies of a differential game to the search of a local minimum. Using a minimization routine, we were able to find saddle point strategies for the pursuit-evasion differential game.

Furthermore, we demonstrate that any deviation from the computed saddle point strategy results in poorer performance for that player. Three examples were shown which demonstrate this characteristic of the differential game solution. In one example, both Pursuer and Defender implement the optimal saddle point strategies. Next, The Defender implements the optimal saddle point strategy and the Pursuer deviates and implements the Pure Pursuit guidance law. We see that the performance of the pursuer is reduced as a result of the the Pursuer implementing a suboptimal policy. We observe the terminal range from the Pursuer to the Target at final time is larger when the Pursuer deviates from his optimal saddle point strategy. Finally, we simulate another scenario wherein the Pursuer implements the optimal saddle point strategy and the Defender implements Pure Pursuit. The results of the simulation confirm that the Defender's deviation from his optimal strategy lower his performance, and as a result, the terminal range from the Pursuer to the Target is less at final time.

6 References

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Appendix A: Theorem 1 Proof

Proof: The co-state dynamics from the Hamiltonian (13) are obtained using the necessary conditions for optimality [26]

$$\dot{\mathbf{x}} = \frac{\partial \mathcal{H}}{\partial \lambda} \quad \dot{\lambda} = -\frac{\partial \mathcal{H}}{\partial \mathbf{x}} \quad \mathbf{0} = \frac{\partial \mathcal{H}}{\partial \mathbf{u}_P} \quad \mathbf{0} = \frac{\partial \mathcal{H}}{\partial \mathbf{u}_D} \quad (38)$$

Utilizing the partial of the Hamiltonian with respect to the states, we find that optimal costates have constant value

$$\dot{\lambda} = \mathbf{0} \quad (39)$$

Utilizing the partial of the Hamiltonian with respect to agent control, we obtain the following system of equations

$$\lambda_{z_P} \cos \theta_P - \lambda_{x_P} \cos \psi_P \sin \theta_P - \lambda_{y_P} \sin \psi_P \sin \theta_P = 0 \quad (40)$$

$$\lambda_{y_P} \cos \psi_P \cos \theta_P - \lambda_{x_P} \cos \theta_P \sin \psi_P = 0 \quad (41)$$

$$\lambda_{z_D} \cos \theta_D - \lambda_{x_D} \cos \psi_D \sin \theta_D - \lambda_{y_D} \sin \psi_D \sin \theta_D = 0 \quad (42)$$

$$\lambda_{y_D} \cos \psi_D \cos \theta_D - \lambda_{x_D} \cos \theta_D \sin \psi_D = 0 \quad (43)$$

Utilizing the necessary optimality conditions (40)-(43), we find the optimal headings depend solely upon the costates. Through algebraic manipulation, the optimal states as a function of the costates are

$$\psi_D = \cos^{-1} \left(\lambda_{x_D} / \sqrt{\lambda_{x_D}^2 + \lambda_{y_D}^2} \right) \quad (44)$$

$$\theta_D = \cos^{-1} \left(\frac{\sigma_D}{\sqrt{\lambda_{x_D}^2 + \sigma_D^2}} \right), \quad \sigma_D = \frac{\lambda_{x_D}^2 + \lambda_{y_D}^2}{\sqrt{\lambda_{x_D}^2 + \lambda_{y_D}^2}} \quad (45)$$

$$\psi_P = \cos^{-1} \left(\lambda_{x_P} / \sqrt{\lambda_{x_P}^2 + \lambda_{y_P}^2} \right) \quad (46)$$

$$\theta_P = \cos^{-1} \left(\frac{\sigma_P}{\sqrt{\lambda_{x_P}^2 + \sigma_P^2}} \right), \quad \sigma_P = \frac{\lambda_{x_P}^2 + \lambda_{y_P}^2}{\sqrt{\lambda_{x_P}^2 + \lambda_{y_P}^2}} \quad (47)$$

The costates were shown to be constant in (39) and from (44)-(47) we have the relationship between the optimal costates and the optimal control. Therefore the optimal headings of the pursuer and defender are constant and their trajectories are straight lines. $\square$

Appendix B: Derivation of Apollonius Circle

The interception of two agents, as described in Fig. 6 allows us to prove that the locus of interceptions for the heading of the agent, a , is a circle when it is slower than the agent, d .

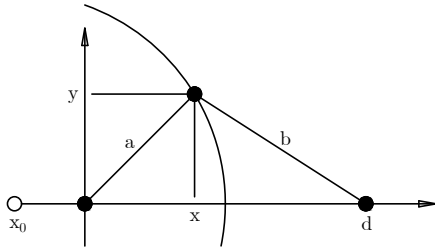


Fig. 6: Interception Geometry of Two Agents

Define the speed ratio of two agents to be γ

$$a = \gamma b \quad (48)$$

Squaring (48), we obtain

$$a^2 = b^2 \gamma^2 \quad (49)$$

Where:

$$a^2 = x^2 + y^2 \quad \text{and} \quad b^2 = (d - x)^2 + y^2 \quad (50)$$

Substituting (50) into (49):

$$x^2 + y^2 = [(d - x)^2 + y^2] \gamma^2 \quad (51)$$

Expanding (51)

$$x^2 + y^2 = x^2 \gamma^2 + y^2 \gamma^2 + d^2 \gamma^2 - 2dx \gamma^2 \quad (52)$$

Bringing terms from (52) to the Left Hand Side and factoring

$$x^2(1 - \gamma^2) + y^2(1 - \gamma^2) - d^2 \gamma^2 + 2dx \gamma^2 = 0 \quad (53)$$

Dividing (53) by $1 - \gamma^2$:

$$x^2 + y^2 - \frac{d^2 \gamma^2}{1 - \gamma^2} + \frac{2dx \gamma^2}{1 - \gamma^2} = 0 \quad (54)$$

Completing the square of (54)

$$\left(x + \frac{d \gamma^2}{1 - \gamma^2} \right)^2 - \frac{d^2 \gamma^4}{(1 - \gamma^2)^2} - \frac{d^2 \gamma^2}{1 - \gamma^2} + y^2 = 0 \quad (55)$$

Bring selected terms from (55) to the Right Hand Side

$$\left(x + \frac{d \gamma^2}{1 - \gamma^2} \right)^2 + y^2 = \frac{d^2 \gamma^4}{(1 - \gamma^2)^2} + \frac{d^2 \gamma^2}{1 - \gamma^2} \quad (56)$$

Therefore, simplifying the Right Hand Side of (56) we have

$$\left(x + \frac{d \gamma^2}{1 - \gamma^2} \right)^2 + y^2 = \left(\frac{d \gamma}{1 - \gamma^2} \right)^2 \quad (57)$$

Let us recall the Apollonius Circle construction outlined in Fig. 7. If we define the Pursuer, P , and the defender, D . we can define the distance between those two agents as

$$d = \overline{PD} \quad (58)$$

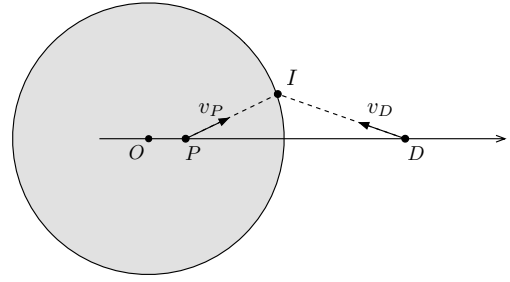


Fig. 7: Apollonius Circle

Furthermore, we define the speed ratio, γ , to be the ratio of the speed of the Pursuer to the Defender

$$\gamma = v_P / v_D \quad (59)$$

Using these definitions, we define the Circle's origin, O , located opposite the Defender on the ray $\overrightarrow{PD}$. We define the distance, $\overline{OP}$ is defined as

$$\frac{\gamma^2 d}{1 - \gamma^2} \quad (60)$$

Furthermore, define the radius of the Apollonius Circle, $\overline{OI}$ as

$$r = \frac{\gamma d}{1 - \gamma^2} \quad (61)$$

Apollonius Circle provides the locus of points describing the intersection of two agents taking a straight-line constant velocity path in a plane. Called Foci, the initial conditions of the two agents described by the points P and D are used to define the circle. Provided a heading of one of the agents, the intersection of the two agents and the corresponding heading of the other agent is found.

One advantage of using the Apollonius Circle, is that it transforms a differential game described by two-agents from one of minimax to a single variable. Provided the heading of one agent, the other can be computed from the intersection point on the circumference of the circle. Both headings describe the minimum length straight-line approach for each agent in a plane.